

Oxygen enrichment

Between 1995 and 1999, the US cement market grew at an average of seven per cent. In response, domestic manufacturers looked at all available means to increase plant capacity. During 1996, Southdown reconsidered the potential of oxygen enrichment reviewing, among other reported data^{1,2,3,4}, the extensive programme of testing conducted by Southwestern Portland Cement during the 1960s. Southwestern was subsequently acquired by Southdown and in 2000, Southdown was acquired by Cemex.

All of the reports described either enrichment of the kiln main burner or riser firing in air-through kiln systems. The general finding was that, although some capacity increase was possible, the difficulty of maintaining an appropriately positioned lance in the kiln, the hazards of increasing flame temperature in the kiln and the cost of liquid oxygen all combined to outweigh the potential benefits. Despite continuing attempts, by the 1990s the use of oxygen was rare if practiced at all.

During Southdown's review, however, it occurred to them that the problems associ-

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In a constant effort to increase plant capacity, Cemex, formerly Southdown, reconsidered data collected in the 1960s regarding the possibilities associated with oxygen enrichment. However, potential hazards and the cost of liquid oxygen meant this process had limited practical implications. Despite this, Southdown believed that these problems could be overcome, making calciner enrichment effective for increased production, particularly for short-term demand. The subsequent testing at the Lyons, CO plant and operation of oxygen enrichment are outlined in this article.

ated with injecting oxygen into the main burner could be avoided if oxygen was fed to the secondary firing point of a precalciner kiln. It was appreciated that, in many cases, a more economic approach would be to upgrade the kiln ID fan. However, the capital cost, permit requirements, equipment lead time, and kiln down-time for installation would prevent rapid response to an immediate need for increased production while, in many cases, calciner vessels are poorly designed and could not easily accommodate an increased draught without a negative impact upon their combustion process. Increased draught may also result in excessive preheater pressure

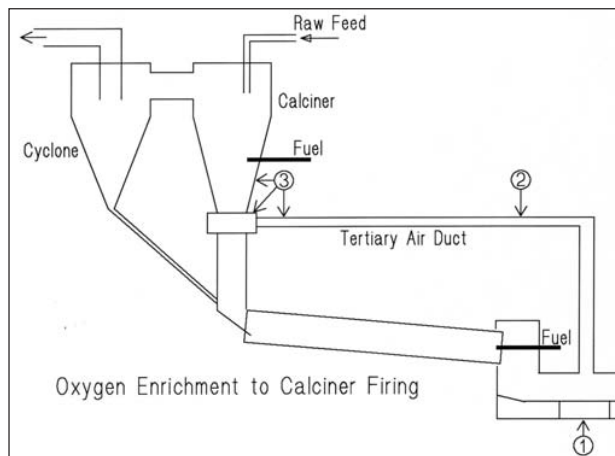
drop or duct velocities. Southdown believed, therefore, that a real benefit might be achieved by calciner enrichment particularly for short-term satisfaction of peak cement demand.

Initial testing was conducted at Southdown's Lyons, CO plant during March 1997. Lyons has a single line, which was originally a long dry kiln constructed by FLSmidth in 1969. Due to the high level of kerogens in the raw materials, the plant experienced unacceptable hydrocarbon emissions. In 1980, Martin Marietta, who then owned the plant, contracted with Fuller Company to convert the kiln to a precalciner system with a single stage pre-

Figure 1: liquid oxygen supply system



Figure 2: oxygen injection points



heater. This unusual configuration allowed the kiln feed to be introduced directly at above 850°C in order to prevent distillation and emission of un-burned hydrocarbons. The calciner is of IHI's SF design with kiln exhaust gas and tertiary air pre-mixed in a swirl chamber. The calciner vessel is small, and it was considered that increasing ID fan capacity would extend combustion into the cyclone and beyond to the exhaust duct. The kiln is coal fired with approximately a 50:50 per cent fuel split between main burner and calciner. Benchmark clinker production was 56tph.

After discussions with several gas suppliers, Air Products was selected to supply cryogenic oxygen. The on-site plant included storage tanks for the liquid gas, evaporators, and a valve train (Figure 1).

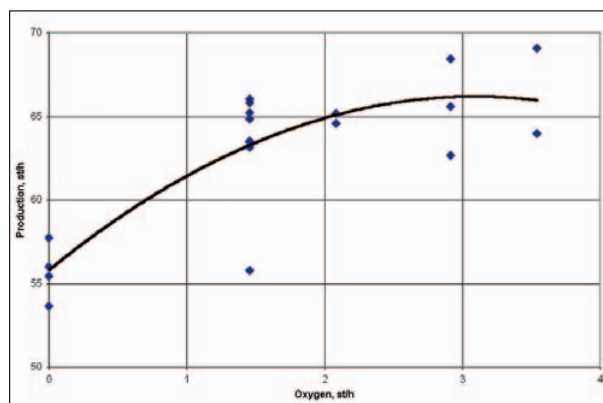


Figure 4: incremental clinker

The literature suggested a typical enrichment rate was two per cent of base kiln draught or 10 per cent of available oxygen.

Initially the oxygen was injected to the tertiary duct above the burner floor. Subsequently, a number of different locations were tried between the tertiary recuperation zone of the clinker cooler and the calciner vessel (Positions 1, 2 and 3 in Figure 2). The location of injection point to the tertiary was not found significantly to vary the effect, and the ultimate position selected was at the swirl chamber of the calciner (Position 3 and Figure 3); this being a less hostile environment than experienced nearer to the cooler take-off. The injector is a simple pipe, more sophisticated diffusers were deemed unnecessary.

Injection of oxygen to the tertiary duct can allow an increased fuel rate to the calciner. In Lyons, however, it was elected to dampen the tertiary to maintain a 50:50



Figure 3: oxygen injection to calciner

per cent fuel split while draught through the kiln was increased. Total fuel and feed rates were increased and the degree of precalcination of hot meal to the kiln remained at 70-75 per cent. No significant change in NOx emissions was detected. The

incremental clinker produced at successive levels of oxygen addition is shown in Figure 4. It is seen that for up to about 2tph or 1.5 per cent of total kiln draught (kiln plus tertiary), some 4.5t of incremental clinker are obtained for each tonne of oxygen. Above this rate, the benefits will decrease.

Actual results may vary somewhat with the fuel efficiency of the kiln system. However, from repeated testing on a number of kilns, Southdown has elected a conservative figure of 3.5t clinker, per tonne oxygen for production reporting. Then, if oxygen costs US\$70/t, the incremental clinker production cost will be US\$20/t plus variable costs. The cost of cryogenic oxygen has been found to vary widely with changes in geographical location and fluctuations in local supply.

Following the successful testing at Lyons, it was decided that the inability of the calciner to accommodate increased draught combined with the high cost of liquid oxygen in Colorado would render an on-site oxygen

generation plant economically attractive. After assessing available systems, Lyons awarded BOC a contract to supply a 50tpd (pure oxygen equivalent) Vacuum Swing Adsorption unit. This plant has been in operation since August 1999 (Figure 5), and it is capable of producing oxygen of 90 per cent purity for a total cost of some US\$20/t (pure oxygen equivalent).

During the past four years, Southdown has engaged in the extensive use of oxygen enrichment on six kilns, including a

KHD precalciner kiln at the Victorville, CA plant. At this plant oxygen has also been successfully injected to the tertiary duct.

Southdown Inc has filed a patent application under the Patent Co-operation Treaty in July 1997⁵. The counterpart of this application in the US and other countries is pending, and it is considered that the addition of enriched oxygen to calciner firing is proprietary to Cemex. □

References:

- (1) Humphries & McEntee; *Oxygen Enrichment of Combustion Air in Rotary Kilns*; PCA R&D Report (1961) M-169.
- (2) Gaydos; *Oxygen Enrichment of Combustion Air in Rotary Kilns*; PCA R&D Report (1965), Vol 7, No 3, p49.
- (3) Wrampe & Rolseth; *The Effect of Oxygen upon the Rotary Kiln's Production and Fuel Efficiency; Theory and Practice*; IEEE Transactions of Industry Applications, (1976), Vol 1A, No 6, p568.
- (4) Leger; *Oxygen Lancing for Production of Cement Clinker*; US Patent #5,572,938, (1996).
- (5) International Patent Application PCT/US97/1323.

Figure 5: VSA oxygen production unit

